

Current Biology

Representational drift in barrel cortex is receptive field dependent

Highlights

- Vibrissal S1 contains sparse populations of broadly tuned touch neurons
- Broadly tuned touch neurons are more stable than narrowly tuned neurons
- Before becoming broadly tuned, touch-responsive neurons are narrowly tuned
- Broadly tuned, highly correlated neurons are more stable than less-correlated neurons

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In brief

Ahmed et al. show that a sparse population of broadly tuned touch-responsive neurons is more stable than other narrowly tuned neurons. These broadly tuned neurons are more synchronously and robustly active over time, suggesting a particularly important population in the primary vibrissal somatosensory cortex for stably representing whisker touch.



Article

Representational drift in barrel cortex is receptive field dependent

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SUMMARY

Cortical populations often exhibit changes in activity even when behavior is stable. How behavioral stability is maintained in the face of such “representational drift” remains unclear. One possibility is that some neurons are more stable than others. We examined whisker touch responses in layers 2–4 of the primary vibrissal somatosensory cortex (vS1) over several weeks in mice stably performing an object detection task with two whiskers. Although the number of touch neurons remained constant, individual neurons changed with time. Touch-responsive neurons with broad receptive fields were more stable than narrowly tuned neurons. Transitions between functional types were non-random: before becoming broadly tuned, unresponsive neurons first passed through a period of narrower tuning. Broadly tuned neurons in layers 2 and 3 with higher pairwise correlations to other touch neurons were more stable than neurons with lower correlations. Thus, a small population of broadly tuned and synchronously active touch neurons exhibits elevated stability and may be particularly important for behavior.

INTRODUCTION

Populations of cortical neurons representing specific sensory,^{1–4} cognitive,^{5–8} or motor⁹ features exhibit a baseline level of change even in the context of behavioral stability. This “representational drift” typically involves changing responses of individual neurons as well as the addition and removal of neurons to and from the responsive population.¹⁰ Such changing populations must nevertheless continue to accurately represent a sensory input or effectively evoke a particular movement.^{11,12} One proposed solution the brain could employ to mitigate this problem is through the presence of a subset of neurons with greater stability.¹³

Broadening of receptive fields is a common consequence of sensory processing. Could this be accompanied by greater stability? In mouse primary visual cortex, neurons that are members of ensembles, or groups of highly correlated neurons, exhibit elevated stability.¹⁴ Cortical columns¹⁵ transform sensory input arriving from the thalamus via a feedforward network from L4 to L2,¹⁶ most prominently resulting in receptive field broadening.^{17–19} In the primary vibrissal somatosensory cortex (vS1), representation of whisker touch transitions from a distributed population of narrowly tuned neurons in layer (L) 4 to a sparser ensemble-based representation consisting of correlated neurons with broader tuning in L2.²⁰ A key function of early sensory cortical processing may therefore be to transition from a more unstable population of narrowly tuned neurons in L4 to a more stable population of broadly tuned neurons in L2, providing a more reliable code for downstream readout.¹³

We ask whether the transition to broader tuning and greater pairwise correlations from L4 to L2 is accompanied by greater long-term stability among touch neurons. We re-analyze volumetric two-photon calcium imaging²¹ data from published experiments²⁰ in which we monitored touch neurons longitudinally in well-trained mice stably performing an object localization task with two whiskers. We find that neurons with narrower tuning, as well as neurons in L4, show lower levels of stability than more broadly tuned superficial neurons. Neurons with greater pairwise correlations to other touch neurons exhibit especially high levels of stability. We also find that functional type transitions are highly structured: to become broadly tuned neurons, non-responsive neurons usually pass through a stage of narrow tuning. Moreover, neurons rarely change preferred touch whisker or direction. Our work shows that vS1 dynamics are highly structured, with broadly tuned touch neurons in L2 showing substantially lower levels of drift, potentially facilitating downstream readout.¹³

RESULTS

Tracking functional types of touch neurons over time

To ask whether representational drift was lower among broadly tuned, superficial neurons, we disaggregated a previously published barrel cortex imaging dataset²⁰ (Table S1) in time. Transgenic mice²² (Ai162 X Slc17a7-Cre) expressing GCaMP6s across all excitatory neurons were trained on a two-whisker object detection task (Figure 1A). On each trial, a pole was presented either within whisking range of the two whiskers or in an out-of-reach position. After pole withdrawal, a sound cued mice to respond by licking one of two lickports. Responses to



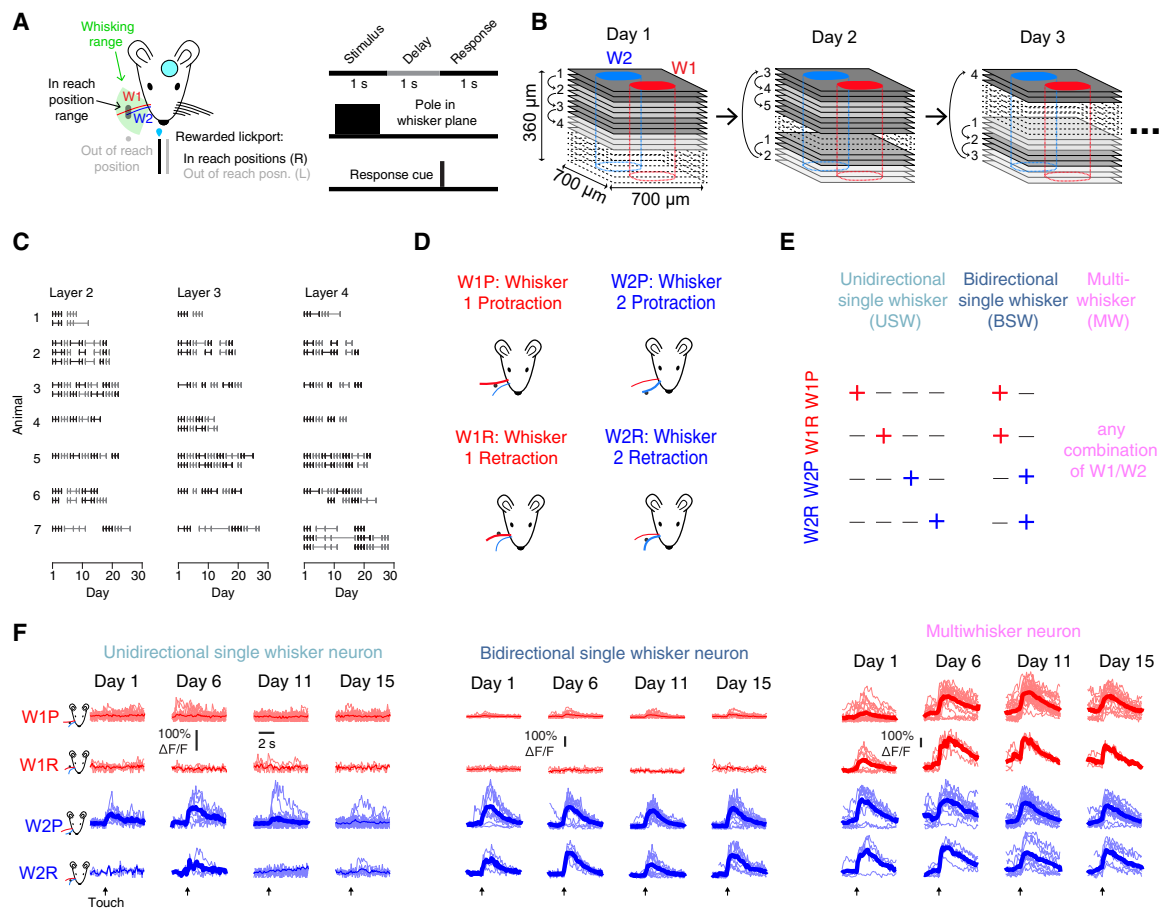


Figure 1. Tracking touch-responsive neurons over many days

(A) Behavioral task. Left, trials with a pole appearing in a range of positions (dark gray) accessible to both whiskers are rewarded on right licks. Trials where the pole appears in a position (light gray) outside the reach of whisking (green) are rewarded on left licks. Right, task timing (STAR Methods).

(B) Imaging strategy. Groups of three planes ("subvolume," same shade of gray) are imaged sequentially on a given day, resuming at the last visited subvolume on the subsequent day. Most but not all subvolumes are typically imaged on any given day.

(C) Merging of data across days for each subvolume in the dataset. Vertical lines indicate days for which data were collected, and horizontal lines indicate aggregated sessions, grouped to ensure sufficient trials for the encoding model (STAR Methods).

(D) Types of touches observed in our task.

(E) Classification of neurons into different touch subpopulations. See also Figures S1 and S2.

(F) Example neuron responses from a single animal from four time points across 2 weeks. Light lines, individual trials. Dark lines, mean. Days with significant response to a particular touch type are shown with a thicker mean line.

See also Table S1 for list of animals used.

the right lickport on accessible pole position trials were rewarded, as were responses to the left lickport on out-of-reach position trials. Once mice reached stable performance (Figure S1A), three 700-by-700- μm planes spaced 20 μm apart in depth were imaged simultaneously. These three planes were dubbed a "subvolume," and each subvolume was imaged for 50–100 trials after which the next subvolume was imaged. We imaged 5–7 subvolumes per mouse (Figure 1B), spanning 300–420 μm total. We started imaging at the layer (L) 1–L2 boundary and continued down to L4. For each layer, subvolumes with at least 750 cells in that layer were used, yielding $2,212 \pm 485$ L2 neurons per subvolume (mean $\pm$ SD, $n = 11$ subvolumes across 7 mice), $1,900 \pm 426$ L3 neurons ($n = 10$ subvolumes across 7 mice), and $2,425 \pm 448$ L4 neurons ($n = 12$ subvolumes across 7 mice). To ensure enough trials for correct

classification of neurons (STAR Methods), we created aggregate sessions for each subvolume from multiple consecutive imaging sessions (Figure 1C), spanning 2.9 ± 1.7 days and 166 ± 22 trials per aggregate session. All analyses were performed using these aggregated sessions. For each analysis, data points for each aggregate session are plotted with respect to the particular 5-day time bin they belong to (days 1–5, 5–10, 10–15, 15–20, or 20–25).

We used high-speed videography (400 Hz, STAR Methods) to detect and classify touches, using a semi-automated algorithm.²³ Touches were classified (Figure 1D) on the basis of the touching whisker (W1 and W2, with W2 more anterior to W1) and the direction of touch (protraction and retraction). Neurons were stably and reliably recorded from on each session across multiple days (Figures S2 and 1C). We used an encoding model

that, given whisker curvature change ($\Delta\kappa$), generated a best-possible predicted $\Delta F/F$, incorporating static curvature change kernel and calcium kinetic kernel (STAR Methods). This model measured how well curvature change, a proxy for follicular force,²⁴ could predict neural activity. Neurons were considered responsive to a specific touch type if on trials with just that touch type, the encoding model's predicted $\Delta F/F$ response Pearson correlation with actual $\Delta F/F$ exceeded 0.15. Neurons that responded to only one whisker-direction combination were considered "unidirectional single-whisker" (USW); neurons responding to both directions for a single whisker, "bidirectional single-whisker" (BSW); and neurons responding to any combination of touches across both whiskers were considered "multi-whisker" (MW; Figure 1E). Despite stable touch counts and task performance (Figure S1), individual neurons showed varying levels of response stability to individual touch types (Figure 1F).

The size of individual touch subpopulations is stable over time

Representational drift typically takes place in the context of a population of neurons whose size is stable.² Therefore, before looking at the stability of individual neurons, we first examined neural stability at the population level (Figure 2A). Across all sessions, the average fraction of touch neurons was small, with USW neurons making up the majority of touch neurons (Figure 2B). Because all subvolumes had data to at least day 10, we compared subpopulation fractions from days 1–5 with 6–10. The fraction of neurons that were touch responsive remained stable over the course of imaging (Figure 2C; L2, paired t test, days 1–5 vs. days 6–10, $p = 0.180$, $n = 11$ subvolumes across 7 mice; L3, $p = 0.347$, $n = 10$ subvolumes across 7 mice; L4, $p = 0.377$, $n = 12$ subvolumes across 7 mice). We next measured the fraction of neurons that belonged to each touch subpopulation over the course of all imaging sessions—USW, BSW, or MW (Figure 2D). Because there were so few MW neurons in layer 4, we excluded these neurons from analysis. The fraction of cells belonging to each subpopulation did not differ between the first 5 days of imaging and the next 5 days of imaging in L2 (Figure 2E; USW, $p = 0.158$; BSW, $p = 0.551$; MW, $p = 0.238$), L3 (USW, $p = 0.416$; BSW, $p = 0.537$; MW, $p = 0.135$), or L4 (USW, $p = 0.390$; BSW, $p = 0.055$). Thus, across layers, the size of specific touch subpopulations was stable over time.

Broadly tuned neurons remain touch responsive for longer periods

Given that we found no change in the aggregate size of the three touch subpopulations, we next compared the stability of individual neurons belonging to these populations. We examined neuronal stability by measuring touch responsiveness over time, grouping touch-responsive neurons belonging to the same touch subpopulation (USW, BSW, or MW) on the first session of imaging. Restricting our analysis to the more numerous protraction touches, we only considered touch responsiveness to W1P and W2P on subsequent sessions and only used neurons that were responsive to protraction on the first imaging day. In L2 and L3, neurons that were USW on the first imaging session showed a larger drop in touch responsiveness at subsequent time points than either BSW or MW neurons (Figures 3A and 3B). This was

also observed when the imaging session that started closest to day 5 was used as baseline (Figure S3A) and when neurons were classified on the last day of imaging (Figure S4A).

To quantify touch responsiveness over time, we measured the average fraction of imaging sessions that neurons belonging to a given touch subpopulation on the first session (USW, BSW, or MW) remained touch responsive. For each neuron in each layer, we calculated the fraction of imaging sessions in which the neuron was responsive to protraction touch (W1P or W2P, calculated separately and averaged; STAR Methods). We then averaged this fraction across neurons belonging to a given subpopulation on the first imaging session. In L2, the fraction of touch-responsive sessions was 0.62 ± 0.06 (mean $\pm$ standard deviation) for USW neurons, 0.83 ± 0.06 in BSW neurons, and 0.90 ± 0.03 in MW neurons (Figure 3C), with broadly tuned neurons exhibiting touch responsiveness for more sessions than more narrowly tuned neurons (USW vs. BSW, paired t test, $p < 0.001$; USW vs. MW, $p < 0.001$; MW vs. BSW, $p = 0.425$). In L3, the fraction of touch-responsive sessions was 0.67 ± 0.06 for neurons that were USW on the first day, 0.82 ± 0.06 in BSW neurons, and 0.88 ± 0.04 in MW neurons. Again, broadly tuned neurons were more stable (USW vs. BSW, $p < 0.001$; USW vs. MW, $p < 0.001$; MW vs. BSW, $p = 0.127$). In L4, the fraction of touch-responsive sessions was 0.59 ± 0.05 for USW neurons and 0.65 ± 0.06 in BSW neurons, and unlike L2 and L3, the difference was not significant (USW vs. BSW, $p = 0.134$). Thus, the fraction of sessions during which each neuron was touch responsive was higher for broadly tuned neurons compared with USW neurons in all but L4. When neurons were classified as belonging to a subpopulation on the fifth day of imaging, broadly tuned neurons were still more stable in layer 2 and layer 3 (Figure S3B). A similar trend was seen when neurons were classified on the last day of imaging (Figures S4B and S4C). Unidirectional single-whisker neurons were comparably stable across layers. Bidirectional single-whisker neurons in L4 were less stable than those in L2 or L3 (L2 vs. L4, unpaired t test, $p = 0.002$; L3 vs. L4, $p = 0.004$).

We next measured the fraction of neurons of each touch subpopulation that, for any given session, remained touch responsive on the subsequent session (Figure 3D). In L2, the average fraction of neurons across animals that remained touch responsive on consecutive sessions was higher in BSW neurons (0.70 ± 0.04 ; mean $\pm$ standard deviation) and MW neurons (0.69 ± 0.03) compared with USW neurons (0.60 ± 0.06 ; paired t tests: USW vs. BSW, $p < 0.001$; USW vs. MW, $p < 0.001$). Similar trends held in L3 (USW: 0.48 ± 0.04 , BSW: 0.67 ± 0.05 , MW: 0.63 ± 0.03 ; USW vs. BSW, $p < 0.001$; USW vs. MW, $p < 0.001$) but not L4 (USW: 0.38 ± 0.06 , BSW: 0.44 ± 0.08). Again, USW neurons were comparably stable across layers, whereas BSW neurons were more stable in L2 and L3 than L4 (unpaired t test; L2 vs. L4, $p < 0.001$; L3 vs. L4, $p = 0.001$).

Among MW neurons, those responding to all four basic touch types were the most likely to be touch responsive on the subsequent session (Figure S5A). Among BSW neurons, neurons responsive to the more anterior whisker exhibited greater touch responsiveness on subsequent sessions (Figure S5B). Among USW neurons, retraction-preferring neurons exhibited greater responsiveness on subsequent sessions than protraction-preferring neurons (Figure S5C). We additionally measured

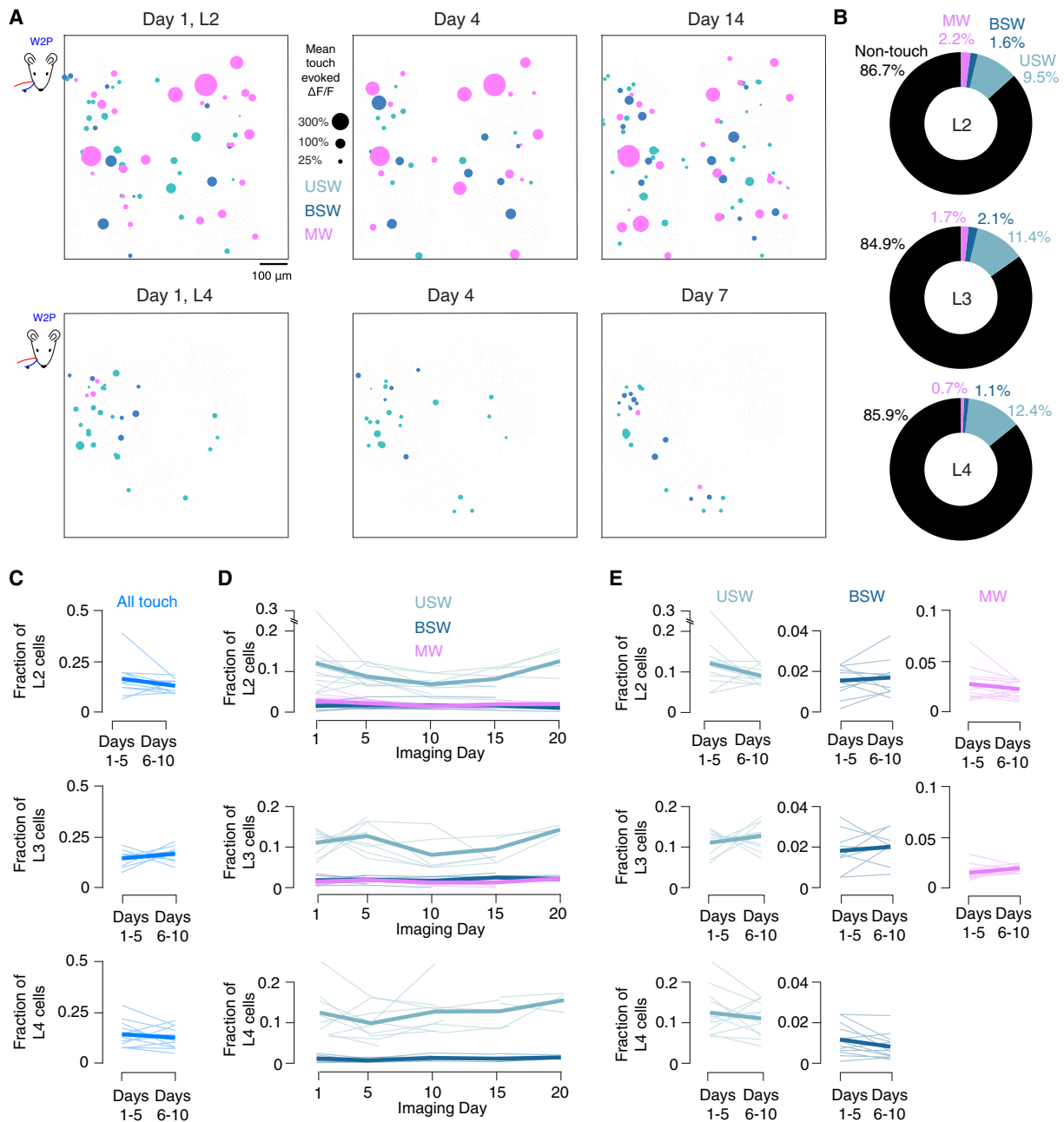


Figure 2. The size of vibrissal representations remains stable over time

(A) Example map showing W2P touch-evoked $\Delta F/F$ for unidirectional single-whisker, bidirectional single-whisker, and multi-whisker neurons across three time points for example L2 (top) and L4 (bottom) subvolumes, projected into one plane from three. Size of circle indicates response magnitude (% $\Delta F/F$) and color indicates subpopulation membership. Light gray points indicate locations of non-responsive neurons.

(B) Touch cell subpopulation composition of each layer averaged across all subvolumes and time points.

(C) Average fraction of neurons responding to touch on aggregate sessions beginning on days 1–5 and 6–10 for each subvolume in each layer (within-subvolume paired t test $p > 0.05$ in all cases). Thin line, individual subvolume. Thick line, cross-subvolume mean.

(D) Fraction of neurons in each layer that are unidirectional single-whisker, bidirectional single-whisker, and multi-whisker across all time points for each subvolume in each layer. Aggregate sessions were grouped for days 1–5, 5–10, 10–15, and 20–25, with each multi-day aggregate session assigned to the bin of its first day. Thin line, individual subvolume. Thick line, cross-subvolume mean.

(E) Average fraction of neurons belonging to a given touch subpopulation on aggregate sessions beginning on days 1–5 and 6–10 for each subvolume in each layer (within-subvolume paired t test $p > 0.05$ in all cases).

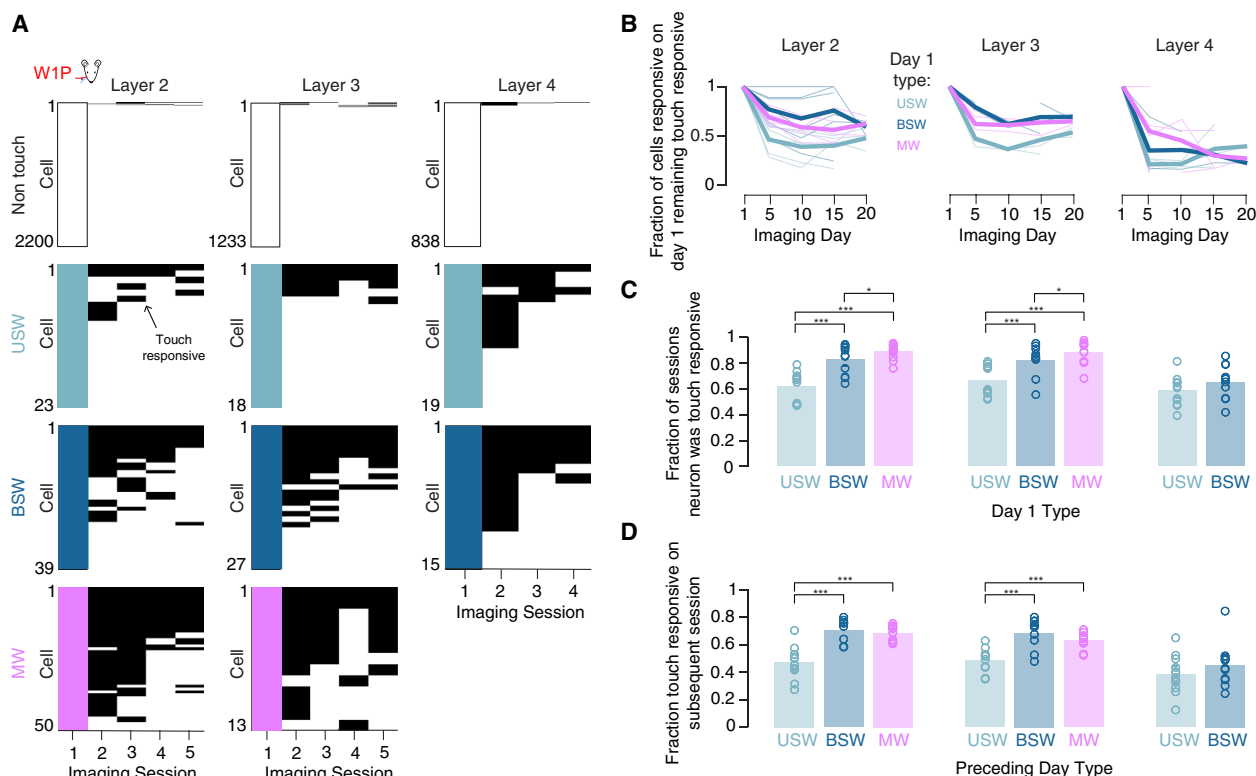


Figure 3. Broadly tuned touch neurons are more likely to remain touch responsive over time

(A) Responsiveness to W1P touches of all neurons over time, for example, subvolume from each layer in a single mouse. Black indicates responsiveness on subsequent sessions. Cells are sorted by subpopulation membership on first session and, within subpopulation, on the number of sessions the neuron was touch responsive. Cells that are unresponsive to touch on the first session are shown in white.

(B) Fraction of cells that are touch responsive at a given time averaged across all mice. Aggregate sessions were grouped for days 1–5, 5–10, 10–15, 15–20, and 20–25, with each multi-day aggregate session assigned to the bin of its first day. Thin line, individual subvolume. Thick line, cross-subvolume mean.

(C) Mean fraction of sessions for which cells of a given subpopulation and layer were touch responsive based on touch subpopulation membership on the first imaging time point. Only cells responsive to W1P or W2P touches on the first session were included, and only responsiveness to W1P or W2P was considered. Bars, mean; circles, individual subvolumes (L2, $n = 11$ subvolumes; L3, $n = 10$; L4, $n = 12$; from 7 mice). p values indicate paired t test, $*p < 0.05$, $**p < 0.01$, $***p < 0.001$.

(D) Mean fraction of cells remaining touch responsive on subsequent time point given classification on preceding time point. Only cells responsive to either W1P or W2P touches on the first session were included.

See also [Figures S3, S4, S5, and S6](#).

touch responsiveness over time in neurons that were whisking responsive compared with neurons that were not responsive to whisking and found no differences in stability over time ([Figures S6A–S6C](#)).

Neurons that were broadly tuned—i.e., MW or BSW—were thus more consistently touch responsive on subsequent sessions than the more narrowly tuned USW neurons, with greater stability in L2/3 than in L4.

Responses of individual broadly tuned touch neurons are more stable over time

Neurons could remain touch responsive over time but nevertheless show a high degree of variability in responsiveness. To quantify neuronal stability more granularly, we computed the correlation of encoding model scores between consecutive imaging session pairs ([Figure 4A](#)). We computed separate correlations for encoding scores on each protraction touch type (W1P and W2P), averaging for each neuron across the touch type(s) to which it was responsive (i.e., if a neuron responded to W1P

and W2P, the average of the two correlations was used, whereas a single correlation value was used for neurons responsive to only W1P or W2P; [STAR Methods](#)). Broadly tuned neurons exhibited greater stability than more narrowly tuned neurons, with a decline over time ([Figure 4B](#)). These trends were also observed when neurons were classified based on their type on the fifth imaging day ([Figure S3C](#)) and when neurons were classified on the last imaging day ([Figure S4D](#)).

We quantified this across layers and touch subpopulations by looking at the mean correlation of encoding scores across consecutive sessions. Correlations were higher in BSW neurons and MW neurons than in USW neurons in L2 ([Figure 4C](#); USW vs. BSW, paired t test, $p = 0.007$; USW vs. MW, $p < 0.001$), L3 (USW vs. BSW, $p = 0.001$; USW vs. MW, $p < 0.001$), and L4 (USW vs. BSW, $p = 0.046$). Encoding score correlations were higher for MW neurons than for BSW neurons in both L2 (MW vs. BSW, $p = 0.037$) and L3 (MW vs. BSW, $p = 0.006$). Encoding score stability was greater in L2 and L3 than L4 for both USW (L2 vs. L4, unpaired t test, $p = 0.001$; L3 vs. L4, $p = 0.030$) and BSW neurons

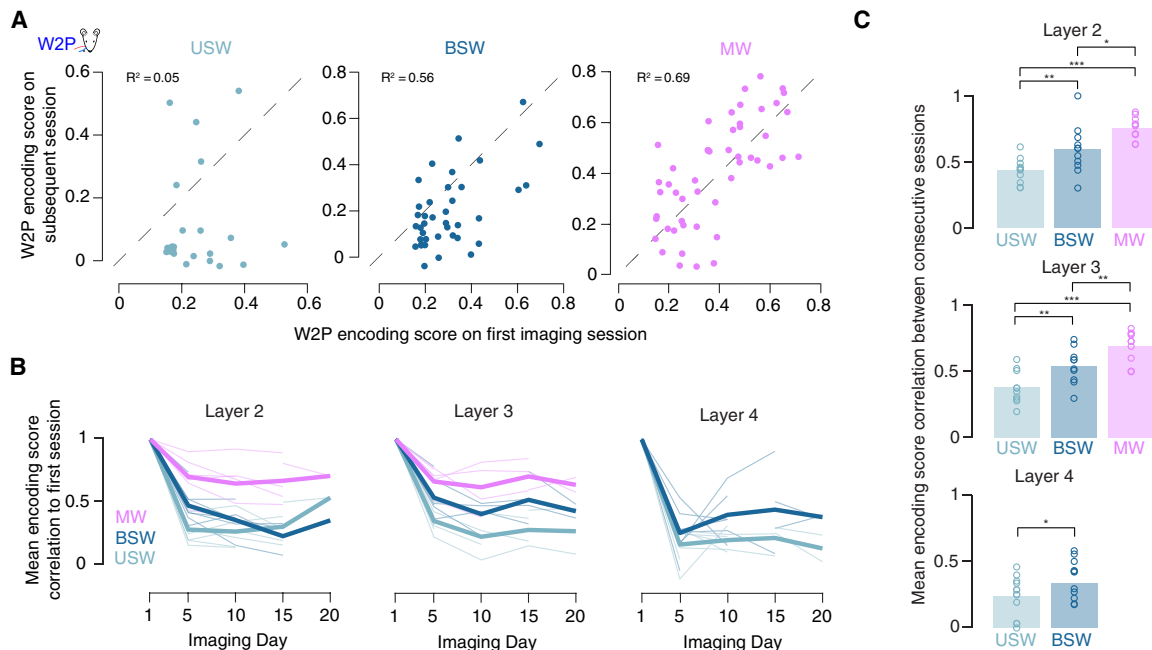


Figure 4. Broadly tuned touch neurons exhibit more consistent responses across time

(A) Encoding scores for W2P touches on the first vs. second session for an example mouse for all layer 2 neurons of a given subpopulation that responded to W2P. R² indicates Pearson correlation.

(B) Correlation between encoding score on the first session and encoding score on all subsequent sessions. Thin lines, individual subvolumes. Thick lines, mean across subvolumes. Correlations for all touch types for which a neuron was responsive to on day 1 are averaged for subsequent days. Aggregate sessions were grouped for days 1–5, 5–10, 10–15, and 20–25, with each multi-day aggregate session assigned to the bin of its first day.

(C) Average Pearson correlation of encoding scores between sessions for all neurons of a given subpopulation and layer, averaged across touch types to which a neuron was responsive. Bars, mean; circles, individual subvolumes (L2, *n* = 11 subvolumes; L3, *n* = 10; L4, *n* = 12; from 7 mice). *p* values indicated for paired *t* tests, **p* < 0.05, ***p* < 0.01, ****p* < 0.001.

See also [Figures S3, S4, S6, and S7](#).

(L2 vs. L4, *p* = 0.003, unpaired *t* test; L3 vs. L4, *p* = 0.008). These trends were also observed in L2 and L3 when neurons were classified based on their type on the fifth imaging day ([Figure S3D](#); L2, USW vs. BSW, paired *t* test, *p* = 0.005; USW vs. MW, *p* < 0.001; L3, USW vs. BSW, *p* = 0.130; USW vs. MW, *p* < 0.001) and when neurons were classified on the last imaging day ([Figure S4E](#)).

Among MW neurons, encoding score stability was lower for neurons responding to more touch types ([Figure S7A](#)). Encoding score stability was comparable across both whiskers for BSW neurons ([Figure S7B](#)). Unidirectional whisker neurons showed comparable levels of stability across preferred touch types ([Figure S7C](#)). We additionally measured encoding score stability over time in neurons that were whisking responsive compared with neurons that were not responsive to whisking and found no differences in stability over time ([Figures S6D and S6E](#)). In sum, encoding model scores exhibited higher session-to-session stability in more broadly tuned cells across all layers, and L4 exhibited lower encoding score stability than either L2 or L3.

Transitions between touch subpopulations are not randomly distributed

We next examined the relative stability of subpopulation membership for individual neurons, looking both at what neurons

belonging to a given subpopulation became on subsequent sessions ([Figure 5A](#)) and which subpopulation a neuron belonged to in prior sessions ([Figure 5B](#)). We quantified stability by calculating the frequency with which neurons changed touch subpopulation²⁵ ([Figure 5C](#)). Neurons that were non-touch responsive or USW on a given session were most likely to have been non-touch responsive on the prior session. This was true in all three layers examined (USW probability of having been non-touch on preceding session, L2: 0.58 ± 0.09; L3: 0.59 ± 0.10; L4: 0.69 ± 0.09; non-touch probability of having been non-touch on preceding session, L2: 0.92 ± 0.03; L3: 0.91 ± 0.02; L4: 0.90 ± 0.03).

By contrast, for broadly tuned neurons (BSW and MW neurons), the raw transition frequencies were highest for transitions to and from the same subpopulation. That is, across all layers, BSW neurons had a higher probability of having been bidirectional in the session before and remaining bidirectional the given session than belonging to any other subpopulation. Similarly, MW neurons in L2 and L3 had a higher probability of having been MW in the session before and remaining MW in the session after than of belonging to any other subpopulation. Moreover, BSW neurons had a higher likelihood of having been touch neurons with narrower tuning on the preceding session than of having been non-touch (BSW probability of being non-touch on preceding session, L2: 0.16 ± 0.09; L3: 0.21 ± 0.19; BSW probability

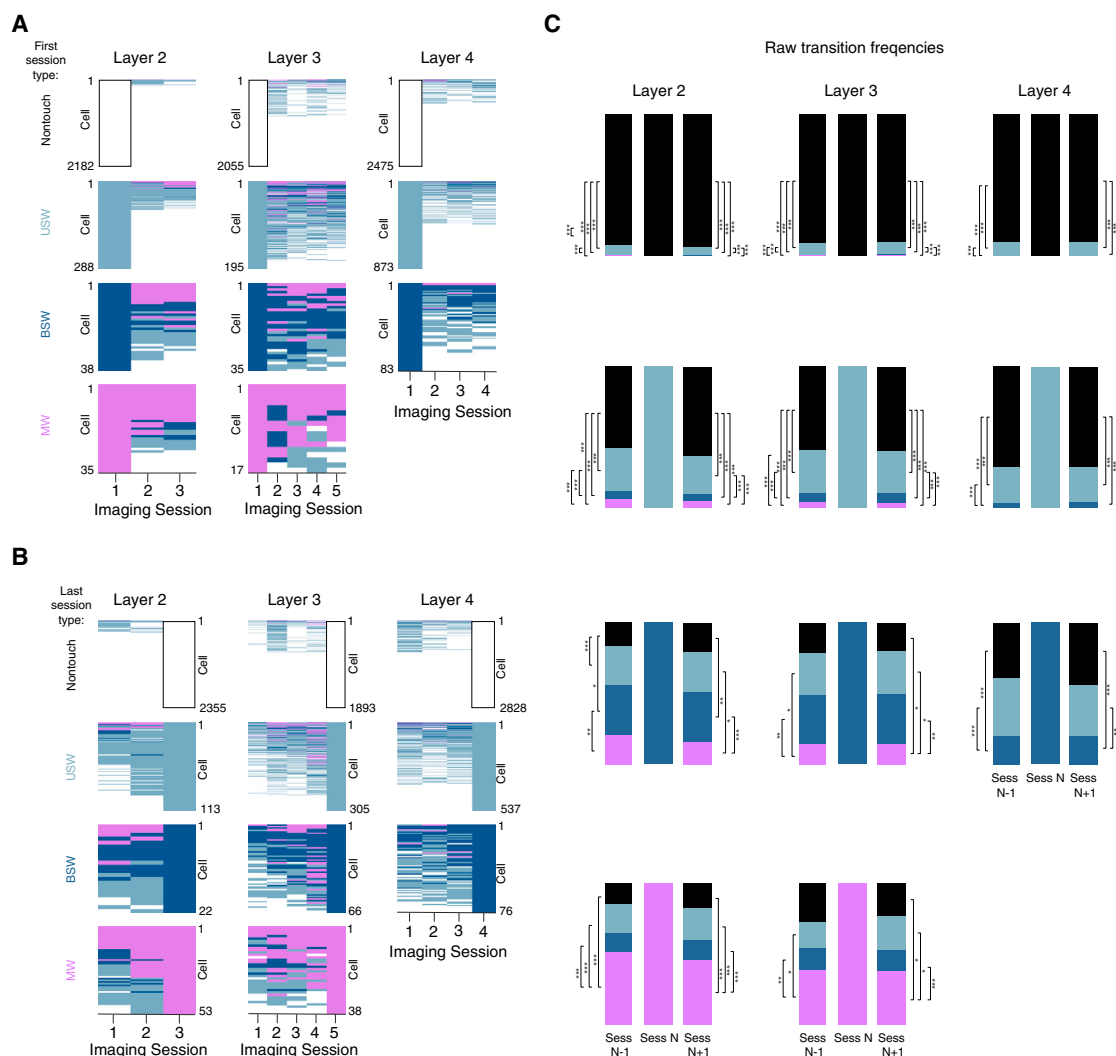


Figure 5. Transitions between neuronal subpopulations are not randomly distributed

(A) Touch subpopulation membership for neurons in example subvolume for each layer, sorted by touch subpopulation (or non-touch, white) on first imaging session. For each cell, its functional type on every subsequent imaged session is indicated. Absence of color indicates non-touch sessions and neurons. Neurons within a subpopulation are sorted by responsiveness.

(B) As in (A) but grouped based on touch subpopulation on final imaging session.

(C) Transition frequencies to and from each neuronal subpopulation for all imaged layers, averaged across subvolumes, sessions, and animals. *p* values indicated for post-hoc Bonferroni correction following ANOVA comparing composition of all subvolumes within a layer; **p* < 0.05, ***p* < 0.01, ****p* < 0.001.

See also Figures S8, S9, and S10.

of being USW touch: L2: 0.27 ± 0.08 ; L3: 0.30 ± 0.07 ; non-touch vs. USW touch, paired *t* test, L2: *p* = 0.039; L3: *p* = 0.646). MW neurons were also far more likely to have been touch cells with narrower tuning (BSW or USW) than of having been non-touch in both L2 and L3 (MW probability of being non-touch on preceding session, L2: 0.15 ± 0.06 ; L3: 0.27 ± 0.25 ; MW probability of being USW or BSW touch: L2: 0.34 ± 0.07 ; L3: 0.34 ± 0.09 ; non-touch vs. narrower touch, paired *t* test, L2: *p* < 0.001; L3: *p* < 0.001). This implies that the sequence of functional types progresses with higher probability along specific trajectories, with broadly tuned neurons first passing through periods of narrower tuning.

Individual touch subpopulations were different in size (Figure 2C), and the raw transition probabilities did not account for this. To assess what transition frequencies look like relative to the null model of random transitions between subpopulations, we normalized the raw transition frequencies to the size of each subpopulation (STAR Methods). For all subpopulations except for USW neurons, normalized transition frequencies were highest for transitions to and from the same subpopulation (Figure S8). Normalized transition frequencies for USW neurons were evenly split between transitions to and from USW, BSW, and MW subpopulations. When normalized by subpopulation size, broadly tuned neurons were highly stable—both MW and

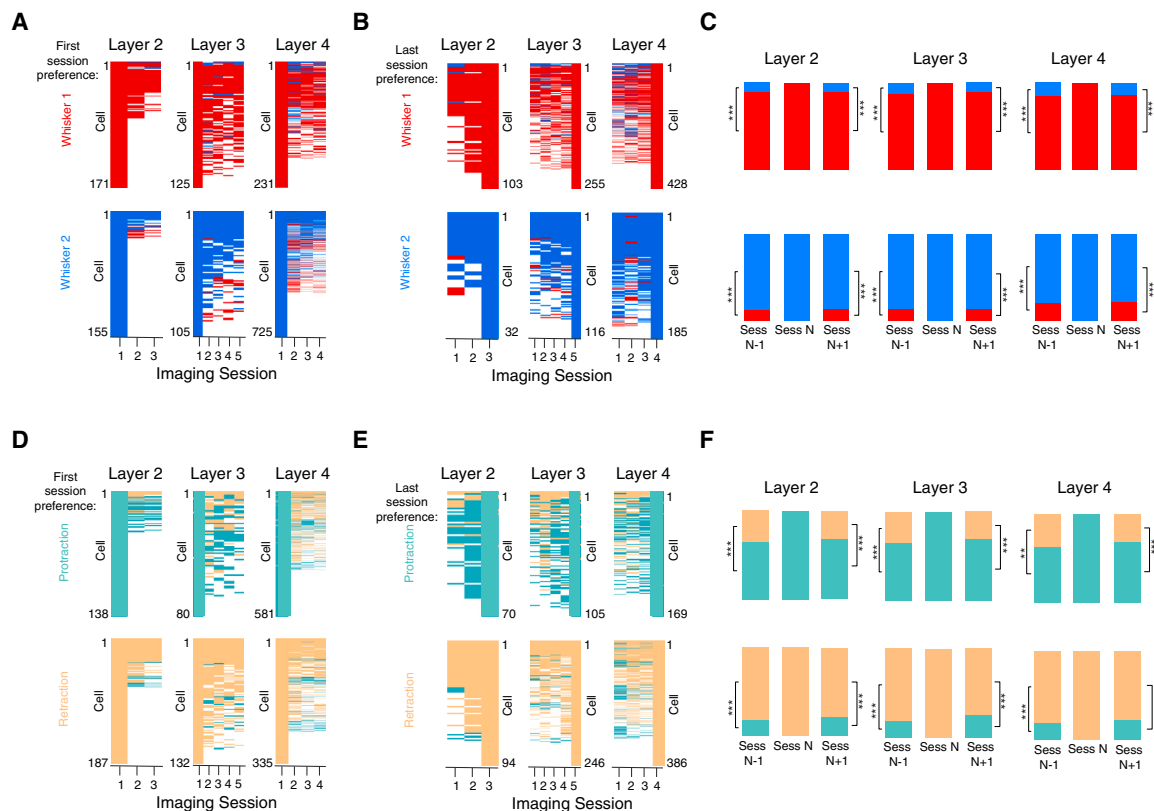


Figure 6. Whisker and directional preferences are stable

(A) Whisker preference for neurons in example subvolumes for each layer, sorted by preference on first imaging session. For each cell, its preference on every subsequent imaged session is indicated. Neurons of a given preference are sorted by touch responsiveness.

(B) As in (A) but based on whisker preference on final imaging session.

(C) Probability of transition within and between whisker preferences. *p* values indicated for within-subvolume paired t test; **p* < 0.05, ***p* < 0.01, ****p* < 0.001.

(D) Directional preference for neurons in an example subvolumes from each layer, sorted by preference on first imaging session. For each cell, its preference on every subsequent imaged session is indicated. Neurons within preference are sorted by responsiveness.

(E) As in (D) but based on directional preference on final imaging session.

(F) Probability of transition within and across directional preference. *p* values indicated for within-subvolume paired t test; **p* < 0.05, ***p* < 0.01, ****p* < 0.001.

BSW neurons were more likely to remain within subpopulation than to switch.

We additionally measured transition frequencies of neurons across more than just one session and found that broadly tuned neurons were still more likely to remain within type across layers and across days (Figure S9A). Whisker preference and directional preference were stable across sessions (Figures S9B and S9C). We similarly looked at transition frequencies of specific subtypes of neurons and found no differences in transition frequencies of particular subtypes within a type (Figure S10).

Whisker preference is stable in L2/3 for neurons preferring the whisker of the barrel they reside in Wang et al.²⁶ Was whisker preference also stable in our task? To address this, we tracked whisker preference of neurons across time, finding that neurons were more likely to retain their first-session whisker preference on subsequent sessions than to shift preference (Figures 6A and 6B). We quantified this using transition probabilities, restricting analysis only to neurons that were touch responsive on both compared sessions (Figure 6C). The probability of retaining whisker preference was high in all layers (probability of

whisker 2 tuned cell being tuned to whisker 2 in preceding session: L2: 0.86 ± 0.12 ; L3: 0.86 ± 0.15 ; L4: 0.78 ± 0.14 ; probability of whisker 1 tuned cell being tuned to whisker 1 in preceding session: L2: 0.89 ± 0.12 ; L3: 0.88 ± 0.09 ; L4: 0.85 ± 0.09) and significantly greater than the probability of switching (paired t tests: probability of remaining tuned to whisker 1 vs. probability of switching: L2: *p* < 0.001; L3: *p* < 0.001; L4: *p* < 0.001; probability of remaining tuned to whisker 2 vs. probability of switching: L2: *p* < 0.001; L3: *p* < 0.001; L4: *p* < 0.001). Did directional preference also remain stable? We examined the directional preference of individual neurons over time, also finding it to be stable in individual animals (Figures 6D and 6E). Across animals, raw transition frequencies (Figure 6F) across directional preferences were low, so that most neurons maintained the same directional preference in addition to whisker preference.

Thus, functional type dynamics for touch neurons are non-random. First, broadly tuned neurons are more likely to remain broadly tuned and are more likely to first pass through a stage as more narrowly tuned neurons. Second, neurons rarely switch whisker or directional preference.

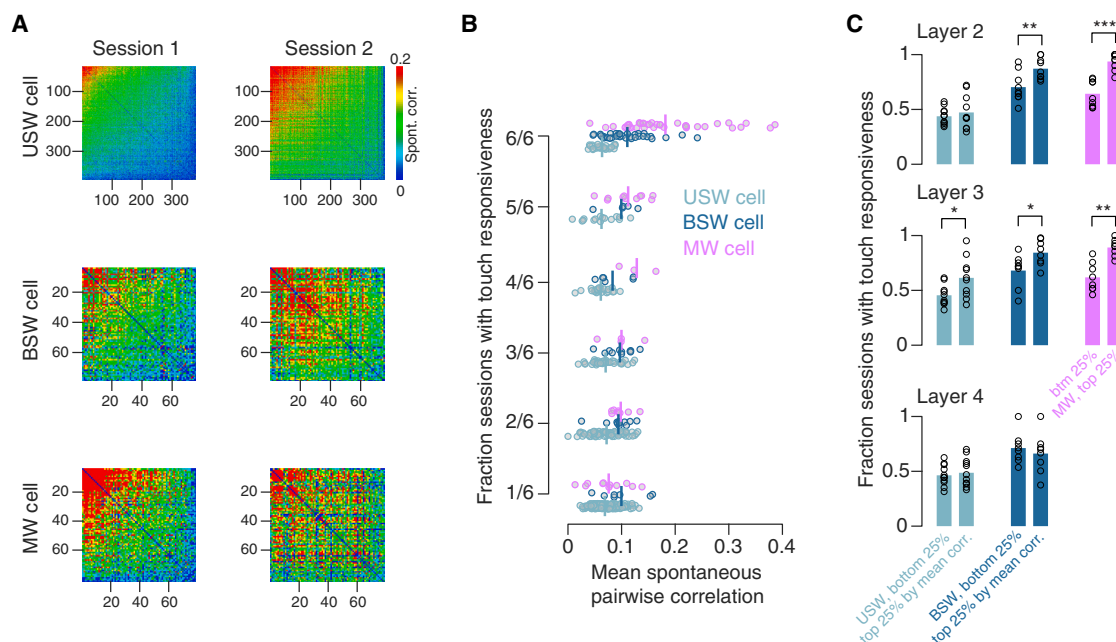


Figure 7. Correlations predict touch responsiveness on subsequent sessions

(A) Pairwise spontaneous activity correlation matrix for an example L2 subvolume for the three touch cell subpopulations on the first and second session. (B) Fraction of sessions with touch responsiveness as a function of mean within-subpopulation pairwise correlation for each neuron in an example L2 subvolume. Color indicates touch subpopulation. Fraction of sessions with responsiveness can only take one of six values because there are only six sessions. (C) Fraction of sessions a neuron is touch responsive as a function of pairwise correlations for three different touch subpopulations across all subvolumes and layers. In all cases, the left bar indicates the neurons having the lowest 25% of pairwise within-subpopulation correlations, and the right bar indicates the most correlated 25%. Bar, mean across subvolumes; circle, individual subvolume. p values indicate paired t test comparing the top with bottom 25%, * $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$.

Neurons with higher pairwise correlations are more stable

In L2/3 of the mouse visual cortex, members of groups of highly correlated neurons exhibit greater long-term stability to both visually evoked and spontaneous activity.¹⁴ Because elevated correlations in spontaneous calcium signals typically predict connectivity,^{27,28} this suggests that stability may be a consequence of circuit structure. Do vS1 neurons exhibit similar correlation-dependent stability? To answer this, we examined pairwise correlations during periods without any whisker touch (spontaneous correlation, STAR Methods) for all touch subpopulations (Figure 7A). Among both MW and BSW neurons, neurons with higher correlations were touch responsive on a greater proportion of sessions (Figure 7B).

We quantified this by partitioning each touch subpopulation (USW, BSW, and MW) into the top and bottom 25% of neurons based on mean pairwise correlation to other subpopulation neurons. We then computed the fraction of sessions for which neurons in these top and bottom quartiles were responsive. Broadly tuned neurons in the top correlation quartile were touch responsive for longer than narrowly tuned neurons in both L2 (BSW, top: 0.87 ± 0.10 , bottom: 0.70 ± 0.14 , top vs. bottom quartile, paired t test, $p = 0.007$, $n = 9$; MW, top: 0.94 ± 0.08 , bottom: 0.64 ± 0.12 , $p < 0.001$, $n = 9$) and L3 (BSW, top: 0.84 ± 0.12 , bottom: 0.68 ± 0.15 , $p = 0.014$, $n = 8$; MW, top: 0.89 ± 0.08 , bottom: 0.62 ± 0.13 , $p = 0.001$). Unidirectional single-whisker neurons only exhibited higher touch responsiveness among more correlated neurons in L3 (USW, top: 0.61 ± 0.19 , bottom: 0.45 ± 0.10 , $p = 0.011$, $n = 10$),

and the difference was smaller than among broadly tuned cells. In L4, correlation did not predict responsiveness (USW, top: 0.49 ± 0.13 , bottom: 0.46 ± 0.09 , $p = 0.285$, $n = 12$; BSW, top: 0.71 ± 0.14 , bottom: 0.66 ± 0.19 , $p = 0.594$, $n = 8$). Thus, in L2 and L3, broadly tuned neurons having higher within-subpopulation correlations were more stable than less correlated neurons. This suggests that stability among broadly tuned neurons may be a consequence of elevated connectivity.^{27,28}

DISCUSSION

We find that broadly tuned vS1 touch neurons—BSW and especially MW cells—exhibit greater stability than more narrowly tuned neurons, with L4 showing lower stability than L2. We find these results despite a sparse population (Figure 2), with MW cells making up less than 5% of the total population size across all sessions. First, the fraction of sessions during which MW and BSW cells remain touch responsive is higher than for USW cells and in L2/3 than in L4 (Figure 3). Second, encoding scores are most stable in MW neurons and in L2/3 and least stable in USW neurons and in L4 (Figure 4). Third, neurons do not move among different touch subpopulations randomly. Instead, MW and BSW neurons are more likely to remain within category, USW neurons are most likely to become non-responsive, and broadly tuned cells typically start out as more narrowly tuned cells (Figure 5). Moreover, neurons rarely switch whisker and directional preference (Figure 6), implying that certain response features are highly stable. Finally, spontaneous pairwise

correlations consistently predict stability of L2/3 but not L4 response within a touch subpopulation over time (Figure 7). Together, our results imply that a small population of broadly tuned, highly correlated touch neurons in L2/3 exhibit elevated stability and may comprise a particularly important population for stably representing whisker touch.

Changes in single-neuron responses have been observed in the face of stable behavior in primary sensory,^{1–4,14,26,29,30} multimodal,⁷ and motor cortical areas^{31,32} as well as hippocampus.^{33,34} Single-neuron responses in vS1 change during whisker-mediated task learning,^{35–37} as well as following whisker trimming.^{38–40} Even when behavior is stable and sensory input is constant, vS1 touch populations exhibit turnover.² We find that broadly tuned neurons—BSW and, especially, MW neurons—are more stable than narrowly tuned single-whisker neurons. From L4 to L2, representations of whisker touch transition from narrow to broad tuning and from distributed to ensemble-based coding.²⁰ This cross-laminar receptive field broadening is found in many sensory cortices.^{17–19,41} Given that neurons participating in ensembles are more stable,¹⁴ a core function of L2/3 may be to transition from less to more stable coding. This is supported by the observation that broadly tuned neurons also dominate vS1 L2/3 output projections to vS2⁴¹ and vM1.⁴² Together, this suggests that less stable neural populations early in the sensory stream funnel activity to more stable populations, which, in turn, disproportionately impact downstream areas. Such a scheme would both counteract some of the problems posed by unstable representations^{11,13} while allowing earlier stages of processing to benefit from the increased flexibility of unstable responses.^{10,43}

Visual cortical neurons did not show layer or areal differences in representational drift,⁴ suggesting that drift is different across modalities. Although broadly tuned neurons project more prominently to both vS2⁴¹ and vM1,⁴² it remains unclear if those areas exhibit greater representational stability. We did not examine intraday drift due to the low number of trials with specific touch trials. In the visual cortex, drift on individual days is largely accounted for by arousal.⁴⁴ Pooling across days to generate aggregate sessions should mitigate against such confounds as we pool from a range of times within individual days, and arousal typically varies systematically in individual behavioral sessions.

We show that specific rules govern touch neuron dynamics. To become broadly tuned, neurons typically first pass through a stage with narrower tuning. Given that we do not record continuously, it is likely that the incidence of such transitions through intermediate states is even higher than observed. One explanation for such dynamics is that neurons change their tuning due to long-term fluctuations in excitability. MW cells in vS1 L2/3 generally show different response amplitudes to various touch types.^{20,45–47} If this non-uniformity in input from individual whiskers is static, then as responsiveness increases, responsiveness to specific touch types will appear in a specific sequence, broadening the receptive field. Excitability is governed by many biophysical properties.^{48,49} If these fluctuate on the relevant time-scale, they could produce the observed dynamics with no changes in underlying synaptic connectivity. Alternatively, fluctuations in excitability could drive plasticity and also produce drift, as observed in hippocampal data.⁵⁰ Finally, synaptic plasticity alone could account for the observed changes, with

elevated connectivity among broadly tuned neurons acting to stabilize their responses,¹⁴ whereas narrowly tuned neurons that lack such stabilizing connectivity exhibit greater instability.

Neurons were unlikely to switch whisker or direction preference. This implies that decoding of touching whisker identity and touch direction is less daunting than it seems: if only 10% of C2 whisker-preferring neurons are responsive at any given moment but the entire potentially responsive population is fixed, a decoder needs to only sample from this C2 superset to ensure consistent readout, regardless of which 10% of neurons are active at any given moment. Consistent with this, touch neurons tuned to their barrel's principal whisker are more stable than neurons tuned to other barrels.²⁶

Broadly tuned neurons with the highest pairwise correlations to the other neurons in their subpopulation exhibited the greatest degree of stability (Figure 7). In mouse visual cortex, a core subset of ensemble neurons exhibits elevated stability during both spontaneous activity and visual stimulation.¹⁴ Correlated spontaneous L2/3 calcium activity predicts connectivity.^{28,51,52} In vS1, declines in touch responsiveness following touch neuron ablation⁵³ and channeling of activity toward natural touch-responsive populations⁵⁴ point to elevated connectivity among touch neurons. Physically, then, a subset of neurons in primary somatosensory cortex exhibiting elevated connectivity and greater long-term stability may act as a scaffold for less stable neural activity. The presence of such “anchor ensembles” would facilitate downstream readout while sampling input from more plastic populations that would benefit from greater flexibility, balancing the benefits of a sparse and stable neural code^{55–57} with those of more flexible representations.^{10,43}

We find that touch responses are more stable among broadly tuned neurons exhibiting high pairwise correlations with other, similar neurons. Stability—along with the presence of these broadly tuned, highly correlated neurons—increases from L4 to L2. Our results suggest that L2/3 functions to stabilize neural activity by transitioning to an ensemble-based coding scheme with broad selectivity.

RESOURCE AVAILABILITY

Lead contact

Further information and requests for resources and reagents should be directed to the lead contact, Simon Peron (speron@gmail.com).

Materials availability

This study did not generate new, unique reagents.

Data and code availability

- Data reported in this paper were previously published.²⁰
- Source code used in this paper has been deposited at [http://github.com/peronlab](https://github.com/peronlab) and is publicly available as of the date of publication.
- Any additional information required to re-analyze the data reported in this work paper is available from the [lead contact](#) upon request.

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AUTHOR CONTRIBUTIONS

A.A., B.V., and S.P. designed the study. B.V. carried out experiments and collected the data. A.A. performed the data analysis. A.A., B.V., and S.P. wrote the manuscript.

DECLARATION OF INTERESTS

The authors declare no competing interests.

STAR★METHODS

Detailed methods are provided in the online version of this paper and include the following:

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SUPPLEMENTAL INFORMATION

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STAR★METHODS

KEY RESOURCES TABLE

REAGENT or RESOURCE	SOURCE	IDENTIFIER
Chemicals, peptides, and recombinant proteins		
Doxycycline (625 mg/kg)	Teklad	TD.01306
Buprenorphine SR	Zoo Pharm	N/A
Ketoprofen (50 mL, 100mg/ml)	Henry Schein	10004031
Experimental models: Organisms/strains		
Transgenic mouse, Ai162	Jackson Labs	JAX 031562
Transgenic mouse, Slc17a7-Cre	Jackson Labs	JAX 023527
Software and algorithms		
Scanimage	Vidrio	Version: 2017
MATLAB	MathWorks	Versions: 2015, 2017
Janelia whisker tracker	Janelia	N/A
Other		
BPod state machine	Sanworks	0.7
Behavioral computer	System 76	Wild Dog Pro
Audio microcontroller	Bela	Bela Starter Kit
Motorized actuators	Zaber	NA11B60-T4-MC04
Microcontroller	Arduino	Uno; Due
Custom lick detection circuit	Janelia Research Campus	N/A
CMOS camera	Basler	Ace-Python 500
Telecentric lens	Edmund Optics	TitanTL
MIMMS two-photon microscope	Janelia Research Campus	N/A
Femtosecond laser	Coherent	Chameleon Ultra 2

EXPERIMENTAL MODEL AND STUDY PARTICIPANT DETAILS

Animals

7 Adult Ai162 (JAX 031562) X Slc17a7-Cre (JAX 023527) mice of mixed sex,²² which express GCaMP6s exclusively in excitatory neurons, were used throughout. To suppress transgene expression during development, breeder mice were fed a diet including doxycycline (625 mg/kg doxycycline; Teklad), so that all experimental mice received doxycycline until weaning. All animal procedures were approved by the New York University Animal Welfare Committee.

METHOD DETAILS

Surgery

For cranial window and headbar implantation, each mouse (8–10 weeks old) was anesthetized with isoflurane (3% induction, 1.5% maintenance). A titanium headbar was attached to the skull with cyanoacrylate (Vetbond). A circular craniotomy (3.5 mm diameter) was made in the left brain hemisphere over vS1 (center: 3.3 mm lateral, 1.7 mm posterior from bregma) using a dental drill (Midwest Tradition, FG1/4 drill bit). After the craniotomy, a double layer cranial window, which was assembled by gluing a 3.5 mm circular #1.5 coverslip to a 4.5 mm circular #1.5 coverslip (Optical Adhesive 61, Norland Products), was placed over the craniotomy. The cranial window and headbar were further affixed to the skull with dental acrylic (Orthojet, Land Dental).

Behavior

Following recovery from surgery, mice were trimmed to two whiskers whose barrels had the least obstructive vasculature, typically C2 and C3, and in some cases, C1 and C2. Mice were placed on water restriction, and subsequent trimming occurred every 2–3 days. Water-restricted mice were head-fixed to the behavioral apparatus and trained on an object localization task,²⁰ in which a metal pole (0.5 mm diameter Drummund Scientific, PA, USA) was vertically moved into the range of the mouse's whiskers either at an out-of-reach position or at a range of accessible proximal positions. On each proximal trial, the pole appeared at a random position drawn from a range spanning 5 mm along the anterior-posterior axis. In all trials, the pole remained accessible for 1–2 s, after which it was

moved downward and out of reach. Pole insertion and removal was accompanied by a 50 ms white noise sound (60–70 dB). 500 ms after the pole was withdrawn, an auditory cue (3.4 kHz, 50 ms) was provided to indicate to the mouse to make a lick response to receive a water reward. Licking the left lickport was rewarded on distal trials, while licking the right lickport was rewarded on proximal trials. On all trials, the lickport remained withdrawn except for during the response epoch (after the auditory cue). Incorrect responses resulted in a timeout (5s) and premature withdrawal of the lickport. Correct trials typically lasted 10 s while incorrect trials lasted 15 s, with mice averaging ~5 trials per minute. Mice were considered to reach criterion performance once d' exceeded 1.5 for two consecutive days, at which point imaging began.

Whisker videography

Whisker video was acquired using custom MATLAB (version 2019a; MathWorks) software from a CMOS camera (Ace-Python 500, Basler) running at 400 Hz and 640 × 352 pixels and using a telecentric lens (TitanTL, Edmund Optics). Illumination was produced via a pulsed 940 nm LED (SL162, Advanced Illumination). 7–8 s of each trial were imaged, covering the 1 s prior to pole movement, the period while the pole was in reach, and several seconds after the pole was withdrawn. Data was processed on the High Performance Computing (HPC) cluster at New York University. First, candidate whiskers were detected with the Janelia Whisker Tracker. Next, whisker identity was refined and assessed across a single session using custom MATLAB software.²³ Following whisker assignment, curvature (κ) and angle (θ) were calculated at specific locations along each whisker's length. Change in curvature, $\Delta\kappa$, was calculated relative to a resting angle-dependent baseline curvature value obtained during periods when the pole was out of reach. Next, automatic touch detection was performed. Touch assignment was manually curated using a custom MATLAB user interface. Protraction touches were assigned negative $\Delta\kappa$ values.

Volumetric two-photon imaging

Two-photon calcium imaging was performed using a custom MIMMS two-photon microscope (<http://openwiki.janelia.org/wiki/display/sharedesigns/MIMMS>), consisting of a 940 nm laser (Chameleon Ultra 2; Coherent) with power rarely exceeding 50 mW. For imaging of each mouse, multiple subvolumes, each consisting of three 700 × 700 μm imaging planes (512 × 512 pixels) which were spaced 20 μm apart, were acquired at a ~7 Hz frequency. On a given imaging day, 4–7 subvolumes were imaged sequentially for 50–70 trials each (Figure 1B). After the first imaging day, motion corrected mean images were collected for each plane and used as reference images during experiments on subsequent days to ensure alignment. Depth was modulated with a piezo (P-725KHDS; Physik Instrumente). Laser power was depth-adjusted with an exponential length constant having a value of 250 μm . Imaging data was acquired using Scanimage (version 2017; Vidrio Technologies).

Imaging data processing

Imaging data was processed on the New York University HPC cluster immediately following acquisition. First, motion correction via image registration was performed. Next, neurons were detected on the first day of imaging using an automated algorithm based on template convolution. The segmentation was verified with post-hoc manual curation. A reference segmentation was thereby established for each plane. On subsequent imaging days, the reference segmentation was algorithmically transferred to the new data.³² Briefly, for each neuron, a 20–25 μm box centered on the neuron was selected and matched via normalized cross-correlation to the subsequent day. Days for which this procedure failed were excluded from analysis, ensuring that only days where the same neurons were imaged were included. Following segmentation, neuropil subtraction²⁰ and $\Delta F/F$ computation²⁰ were performed. The $\Delta F/F$ trace was used for all analyses. For layer assignment, each neuron was first assigned a depth in reference to a single reference plane taken at the top of the dura.²⁰ The L1–L2 border was defined at the depth of the most superficially imaged excitatory neuron and the L3–L4 border was found by manually locating a noticeable shift in neuron morphology in conjunction with the emergence of clearly visible septa. The L2–L3 border was placed at the midpoint between the L1–L2 and L3–L4 borders.

Imaging session aggregation

Data from multiple imaging days was aggregated to obtain at least 150 trials per aggregate (Figure 1C). This ensured that there were sufficient trials to employ the encoding model, which depends on having at least several unique trials per touch type. In figures where data is compared across days, aggregated sessions are plotted based on the first day for that particular aggregate session.

Encoding model and neural subpopulation classification

We used an encoding model to assess how well whisker curvature change could predict neural activity and, therefore, whether a neuron is a touch cell. The model predicts neural activity ($\Delta F/F$), r_{model} , for each neuron using:

$$r_{\text{model}} = [(a_{w_1} + a_{w_2}) * g] + \sigma^2$$

Here, a_{w_i} is the predicted amplitude of response to a given whisker's touch input at a given time, g is the GCaMP kinetics kernel, and σ^2 is a Gaussian noise term.

For single-whisker touch trials for whisker i , a_{w_i} is:

$$a_{w_i} = s_{\text{pro}} \cdot \log_{10}(-\Delta\kappa_{\text{pro}} + o_{\text{pro}}) + s_{\text{ret}} \cdot \log_{10}(\Delta\kappa_{\text{ret}} + o_{\text{ret}})$$

This model is based on previous work using a less constrained generalized linear model that revealed monotonically increasing response as a function of whisker curvature across touch neurons.⁵³ For a given whisker, the amplitude of the response to a protraction touch ($\Delta\kappa < 0$) at a given time, a_{w_i} , is given by applying a slope s_{pro} to its change in curvature, $\Delta\kappa_{pro}$. To account for neurons that have a minimal force needed to elicit a response, the offset term o_{pro} was included. The retraction ($\Delta\kappa > 0$) response is calculated in an analogous manner.

The indicator kinetics kernel, g , consisted of a sum of exponentials having time constants τ_{rise} and τ_{decay} . It was normalized so that its peak was 1. Both τ_{rise} and τ_{decay} were constrained based on the known physiological range⁵⁸: τ_{rise} , 100 ms to 500 ms; τ_{decay} , 1 s to 5 s. The noise term σ^2 was determined for each neuron by measuring the variance of negative $\Delta F/F$ values. Our sliding-window F_0 fitting procedure,² in which we compute F_0 using a 3 minute sliding window as the median for neurons that have low activity (non-skewed F distribution) and the 5th percentile for the most active neurons (highly skewed F distribution) ensures that $\Delta F/F$ is appropriately 0-centered.

The model was fit with 5-fold cross validation using block coordinate descent and a mean-square-error cost function minimizing the difference between model response, r_{model} , and neural response, r_{neural} . During cross-validation, data was partitioned by randomly drawing 5 disjoint equal sized sets of trials; individual trials were not broken up. The terms of a_{w_i} were iteratively fit along with g using single-whisker touch trials and an equal number of non-touch trials. Because this resulted in two estimates of g , we employed the mean of these parameters for the final model fit.

Neurons were classified based on how well the encoding model could predict neural activity on specific trial types.²⁰ Neurons were classified as touch responsive for a given session if the correlation between this predicted $\Delta F/F$ and actual $\Delta F/F$ on a given touch type's trials was greater than 0.15. Thus, neurons were classified as responsive to each of the four touch types (W1P, W1R, W2P, W2R; Figure 1D) independently. Touch responsive neurons were assigned to touch subpopulations based on their responsiveness to combinations of the four touch types – unidirectional single-whisker neurons responded only on one of the four touch trial types, bidirectional single-whisker neurons responded to both retractions and protraction of only a single whisker, and multi-whisker neurons responded to any combinations of touch types that involved both whiskers (Figure 1E). Neurons were classified as whisking responsive based on the correlation between the actual $\Delta F/F$ and predicted $\Delta F/F$ based on whisker angle, θ .⁵³

Touch responsiveness analysis

When performing touch responsiveness analysis over time (Figure 3), only protraction touches were used. For unidirectional single-whisker neurons, we only considered a single touch type; for bidirectional and multi-whisker neurons we averaged the responsiveness for the particular touch types (W1P, W2P) that the neurons was responsive to on the baseline day. Thus, for neurons responsive to both W1P and W2P on day 1, responsiveness could take a value of 0, 0.5, and 1. To produce the averaging across imaging days plot (Figure 3B), we binned across days 1-4, 5-9, 10-14, 15-19 and 20-24, based on the imaging day of the first session in an aggregate session.

Encoding score correlation analysis

For all neurons of a given subpopulation, encoding scores, were calculated for all imaging sessions (Figure 4). For each touch subpopulation (USW, BSW, MW), a vector of encoding scores for both protraction touch types (W1P, W2P) consisting of all neurons that exhibited significant (i.e., greater than 0.15) responsiveness to that touch type was constructed for a given day and the subsequent day (criteria only had to be met on the first of the two compared days). The Pearson correlation was computed for vectors from consecutive days, yielding two correlations per touch subpopulation, one per protraction touch type (W1P, W2P). If a neuron was not significantly responsive on the first day to a particular touch, that value was excluded from both vectors. We then averaged the two correlations together to obtain the overall value for that population for that pair of sessions.

Transition frequency analysis

To analyze the likelihood that a neuron would change tuning from one imaging session to the next, the probability that a neuron of a given subpopulation would transition to or from another subpopulation was calculated. For all neurons that belonged to a specific subpopulation on the first session (non-touch, USW, BSW, MW), the raw transition frequency in the forward direction was calculated as the fraction of neurons that began in a certain subpopulation and ended up in a different subpopulation in the next imaging session (Figure 5). The raw transition frequency in the backward direction was calculated as the fraction of neurons that belonged to a specific subpopulation (USW, BSW, or MW) on the final session and belonged to a different subpopulation in the previous imaging session. To account for differenced in individual touch subpopulation size, raw transition frequencies were normalized to the size of the subpopulation on a given session. Expected transition frequencies were calculated using the size of the neuron population of a given subpopulation on each session. Normalized transition frequencies were calculated by dividing the raw transition frequencies by the expected transition frequencies (Figure S5).

Pairwise correlation versus responsiveness analysis

To assess whether pairwise correlations predicted responsiveness (Figure 7), we first computed pairwise correlations within specific touch subpopulations on the first day of imaging. Specifically, Pearson correlations of $\Delta F/F$ were computed for periods where no touch took place by excluding all time points 1 s prior to and 10 s following a touch. This mostly consisted of trials where the pole was out of reach. For every neuron belonging to any given touch subpopulation (USW, BSW, MW), the mean correlation to all other

neurons in that group was computed. Correlations were sorted and neurons within a population were partitioned into the top and bottom 25% by correlation. Touch responsiveness was assessed for all subsequent sessions as described above.

QUANTIFICATION AND STATISTICAL ANALYSIS

For comparisons across two matched groups, the two-tailed paired t-test was used. Pairing was within-animal. Given the large number of groups in the transition analyses ([Figures 5](#) and [S4](#)), these were compared using a one-way ANOVA with a Bonferroni multiple comparison correction. All statistical analyses were performed using MATLAB, and a list of animals used can be found in [Table S1](#). Statistical tests are additionally described in the results section, and briefly mentioned in relevant figure legends. Parametric statistics were used throughout, though no tests were used to establish normality.